



(12) **United States Patent**  
**Le et al.**

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(54) **BUFFER LAYERS AND INTERLAYERS THAT PROMOTE BiSbx (012) ALLOY ORIENTATION FOR SOT AND MRAM DEVICES**

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This patent is subject to a terminal disclaimer.

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**Related U.S. Application Data**

(63) Continuation of application No. 17/401,856, filed on Aug. 13, 2021, now Pat. No. 11,763,973.

(51) **Int. Cl.**  
**H10N 50/10** (2023.01)  
**G11B 5/127** (2006.01)  
(Continued)

(52) **U.S. Cl.**  
CPC ..... **H10N 50/10** (2023.02); **G11B 5/1278** (2013.01); **G11B 5/235** (2013.01); **G11B 5/314** (2013.01);  
(Continued)

(58) **Field of Classification Search**  
None  
See application file for complete search history.

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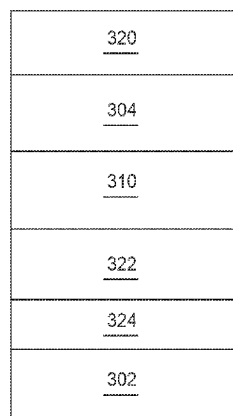
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(57) **ABSTRACT**

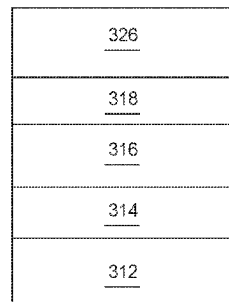
The present disclosure generally relate to spin-orbit torque (SOT) magnetic tunnel junction (MTJ) devices comprising a buffer layer, a bismuth antimony (BiSb) layer having a (012) orientation disposed on the buffer layer, and an interlayer disposed on the BiSb layer. The buffer layer and the interlayer may each independently be a single layer of material or a multilayer of material. The buffer layer and the interlayer each comprise at least one of a covalently bonded amorphous material, a tetragonal (001) material, a tetragonal

(Continued)

301



310, 320



(110) material, a body-centered cubic (bcc) (100) material, a face-centered cubic (fcc) (100) material, a textured bcc (100) material, a textured fcc (100) material, a textured (100) material, or an amorphous metallic material. The buffer layer and the interlayer inhibit antimony (Sb) migration within the BiSb layer and enhance uniformity of the BiSb layer while further promoting the (012) orientation of the BiSb layer.

## 24 Claims, 13 Drawing Sheets

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**G11B 5/235** (2006.01)  
**G11B 5/31** (2006.01)  
**G11B 5/39** (2006.01)  
**H01F 10/30** (2006.01)  
**H01F 10/32** (2006.01)  
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**G11B 5/00** (2006.01)

### (52) U.S. Cl.

CPC ..... **G11B 5/3909** (2013.01); **H01F 10/30**  
 (2013.01); **H01F 10/3254** (2013.01); **H01F**  
**10/329** (2013.01); **H10N 52/80** (2023.02);  
**G11B 2005/0024** (2013.01)

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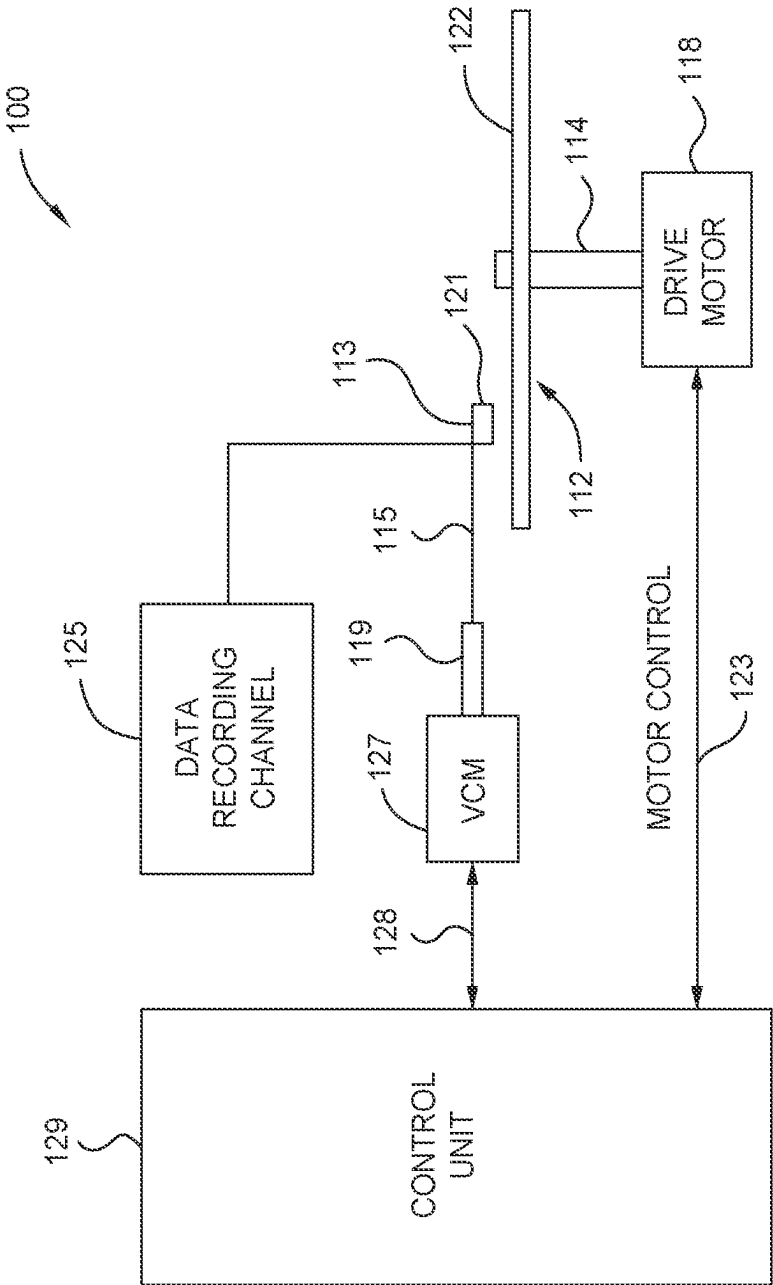
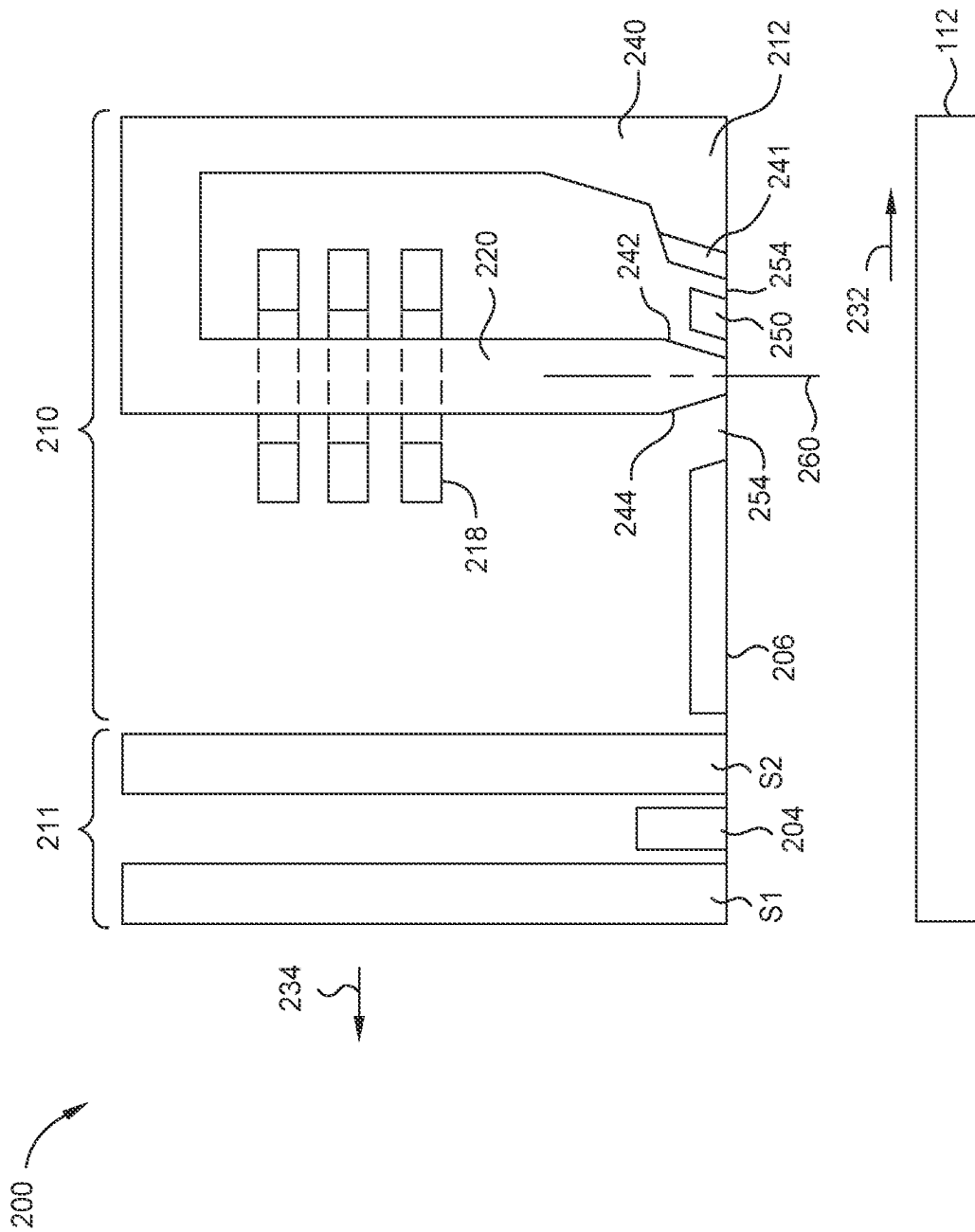


FIG. 1

2. GL

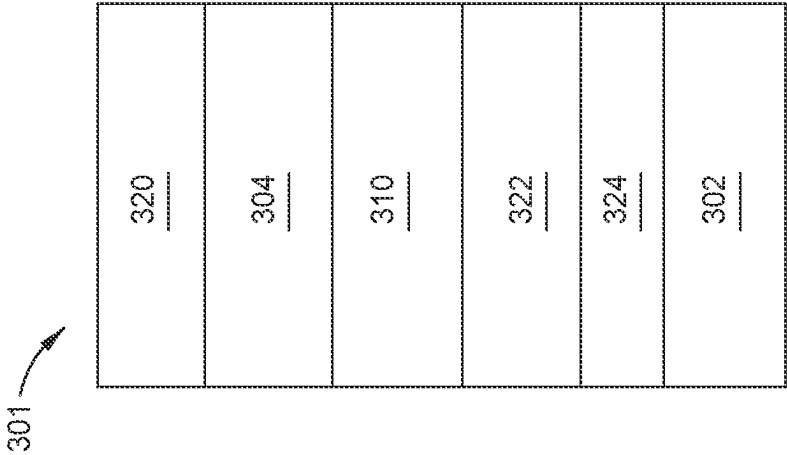


FIG. 3A

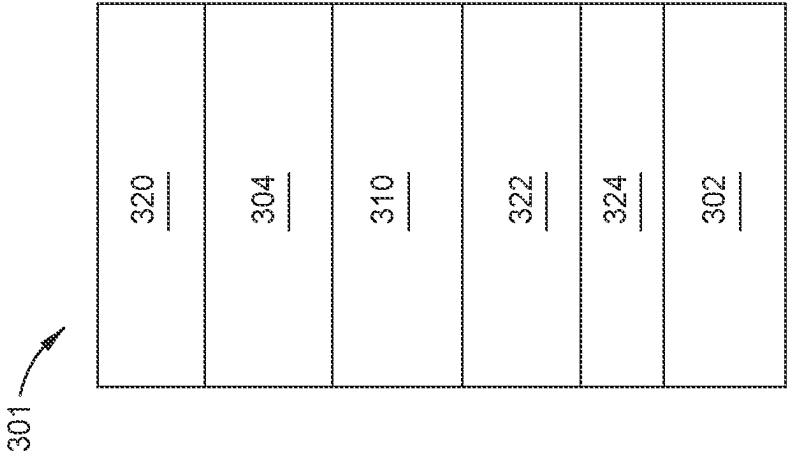


FIG. 3B

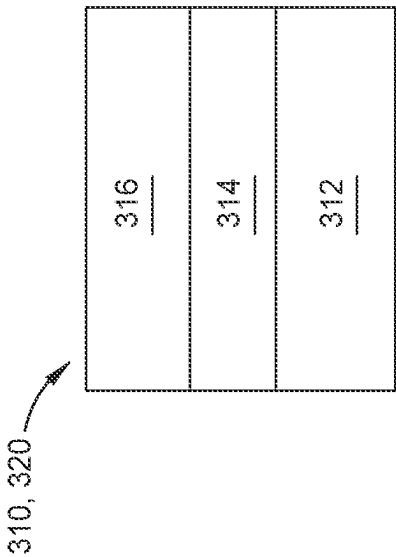


FIG. 4B

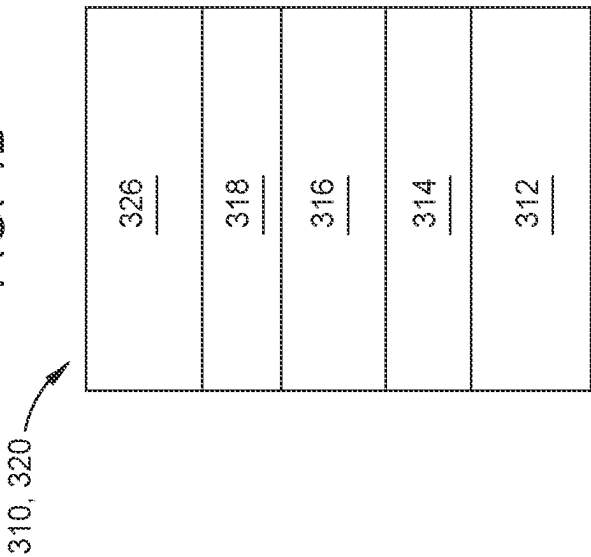


FIG. 4D

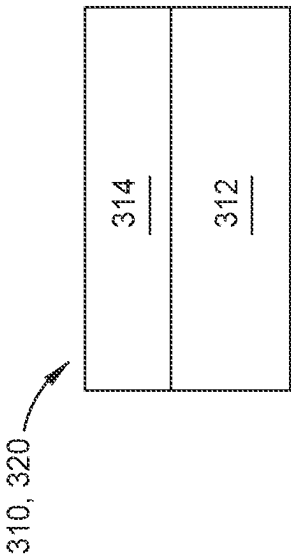


FIG. 4A

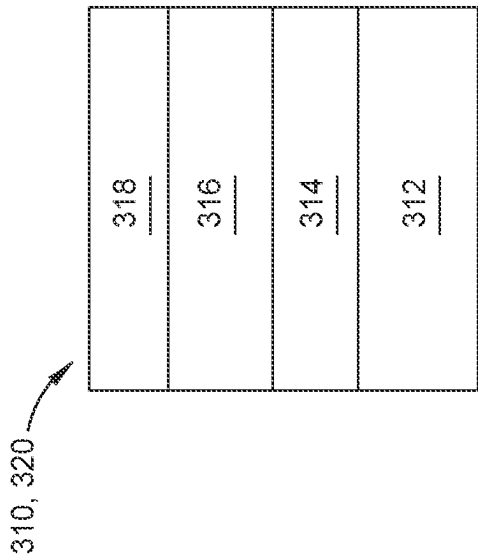
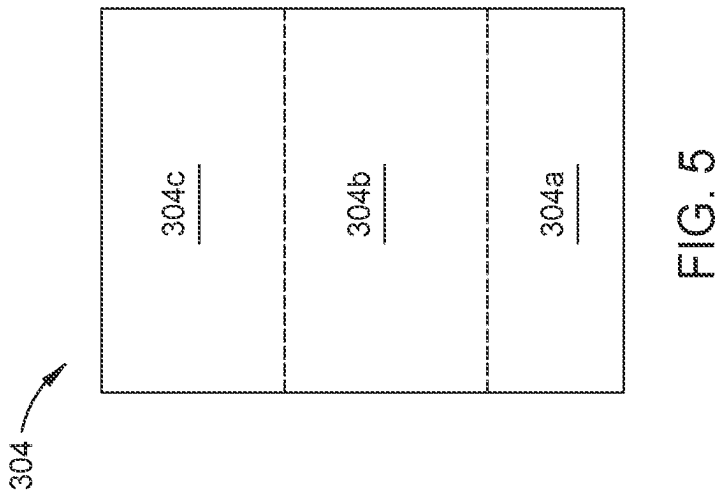


FIG. 4C





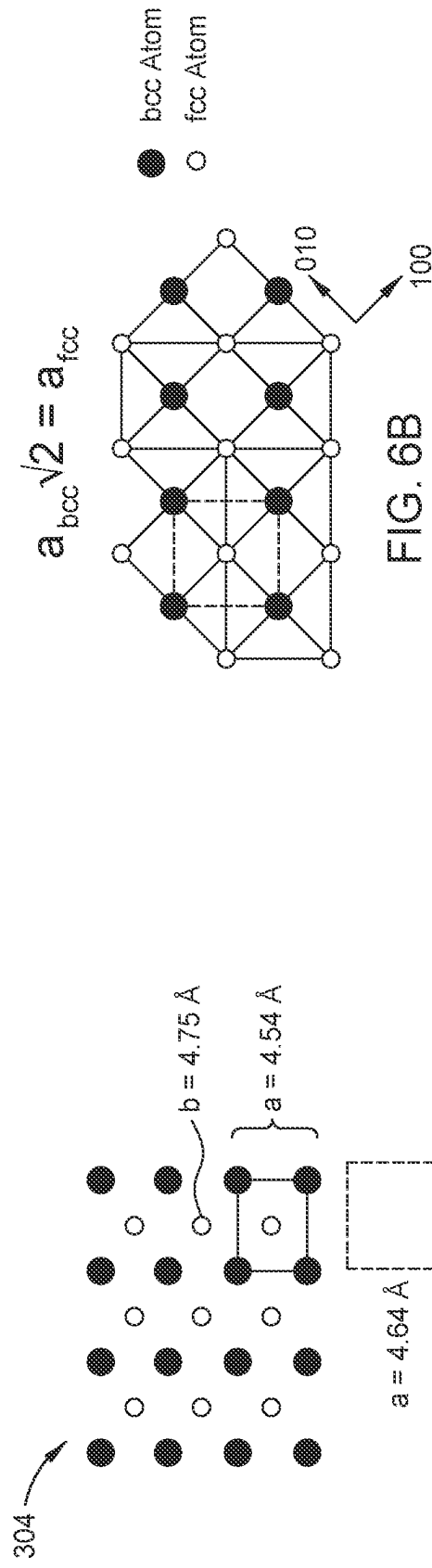


FIG. 6A

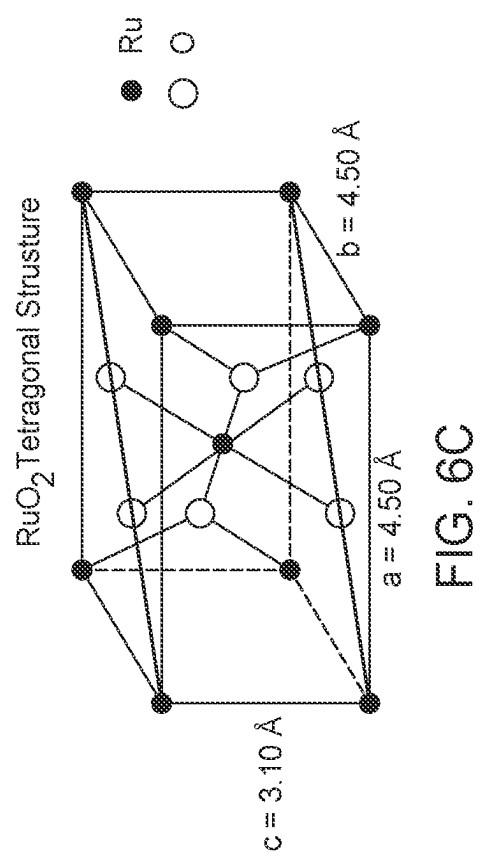


FIG. 6C

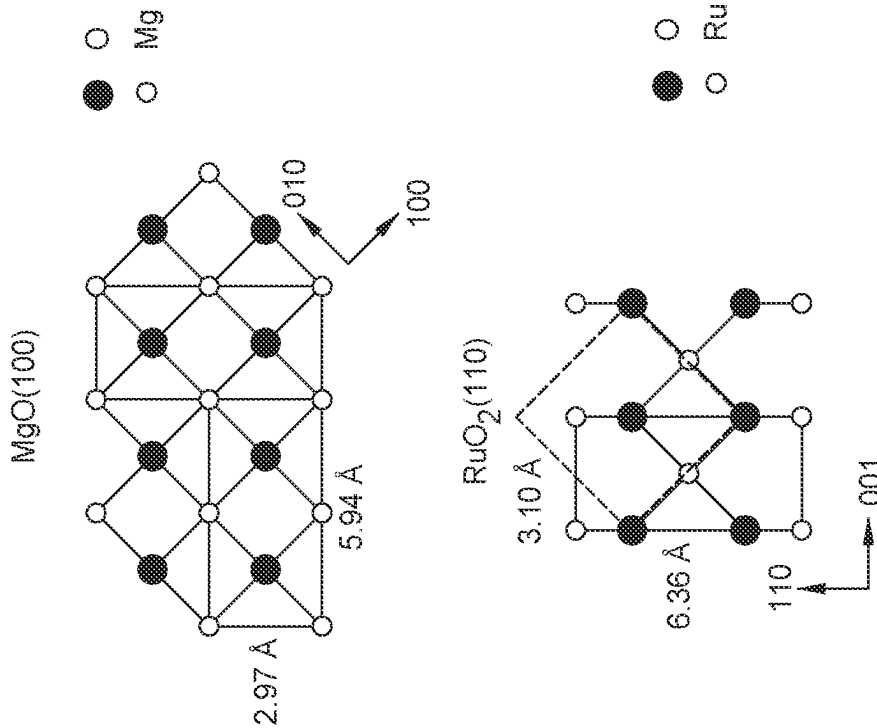


FIG. 6E

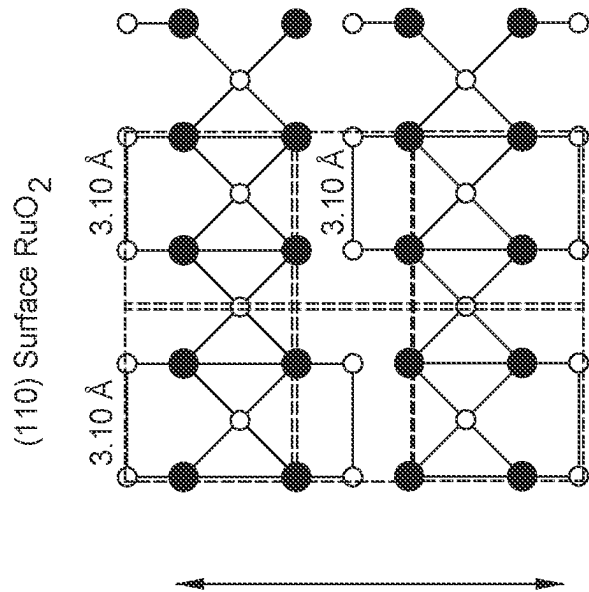


FIG. 6D

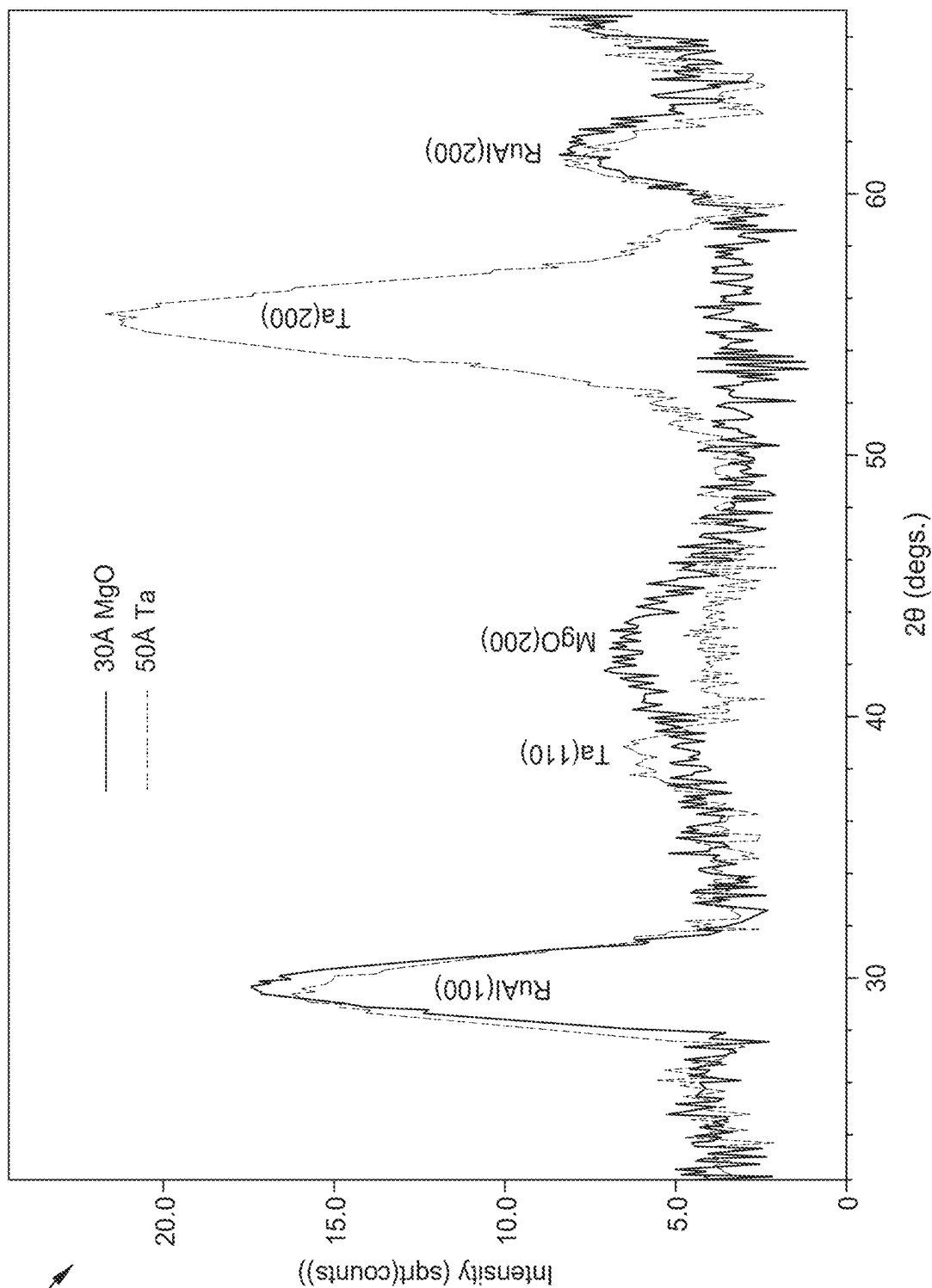


FIG. 7

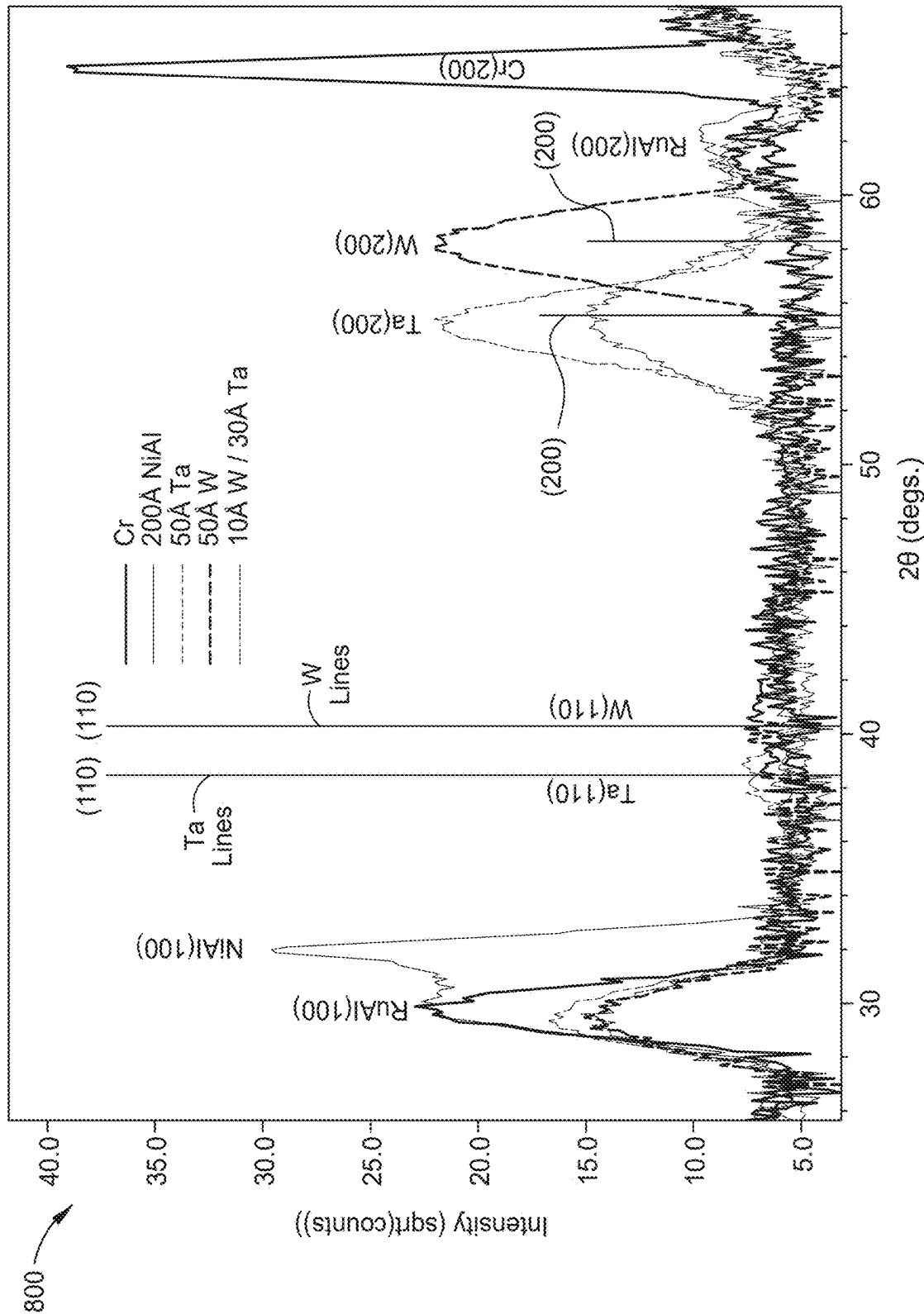


FIG. 8

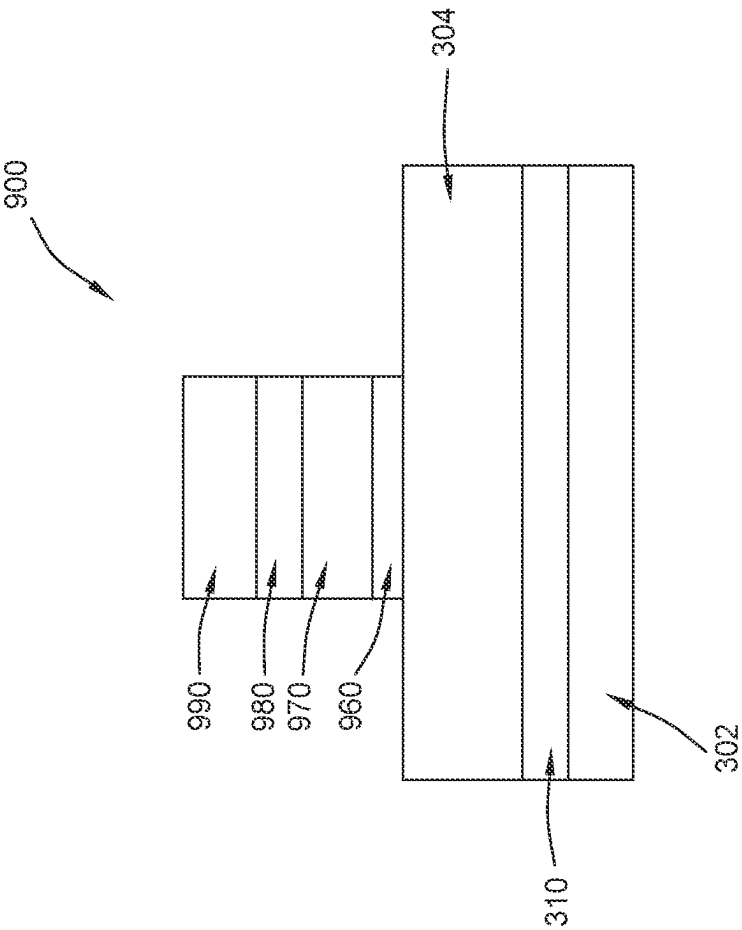


FIG. 9A

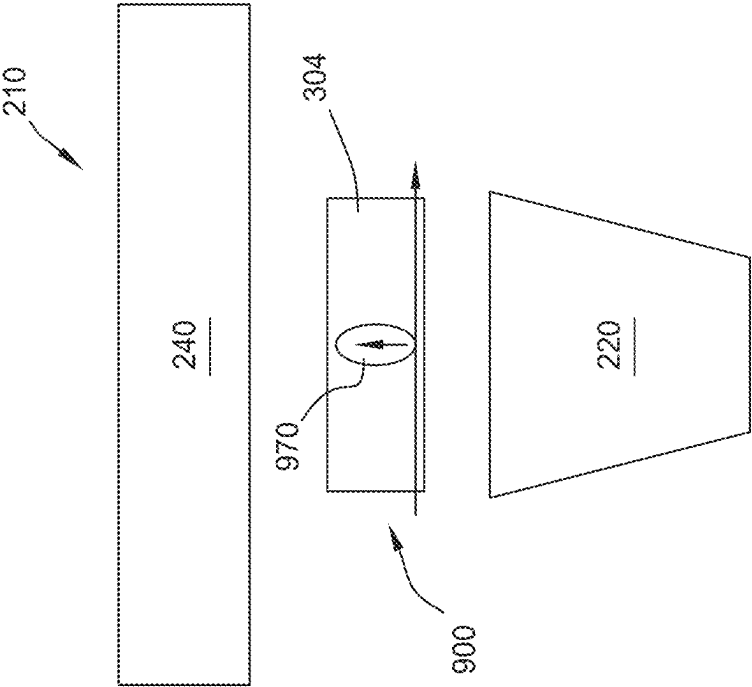


FIG. 9C

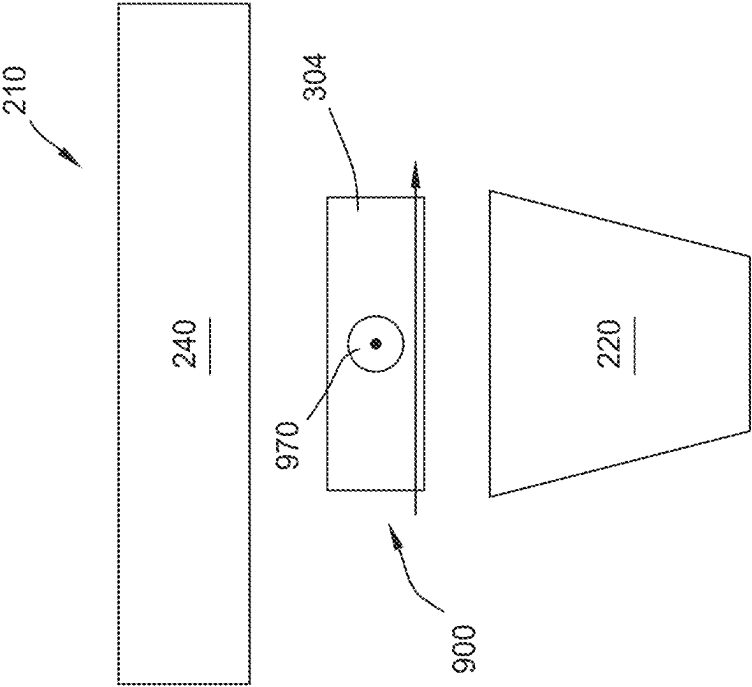


FIG. 9B

1000

1001

310  
1040

|            |  |  |             |             |             |
|------------|--|--|-------------|-------------|-------------|
| <u>304</u> |  |  | <u>1030</u> | <u>1020</u> | <u>1010</u> |
|------------|--|--|-------------|-------------|-------------|

FIG. 10

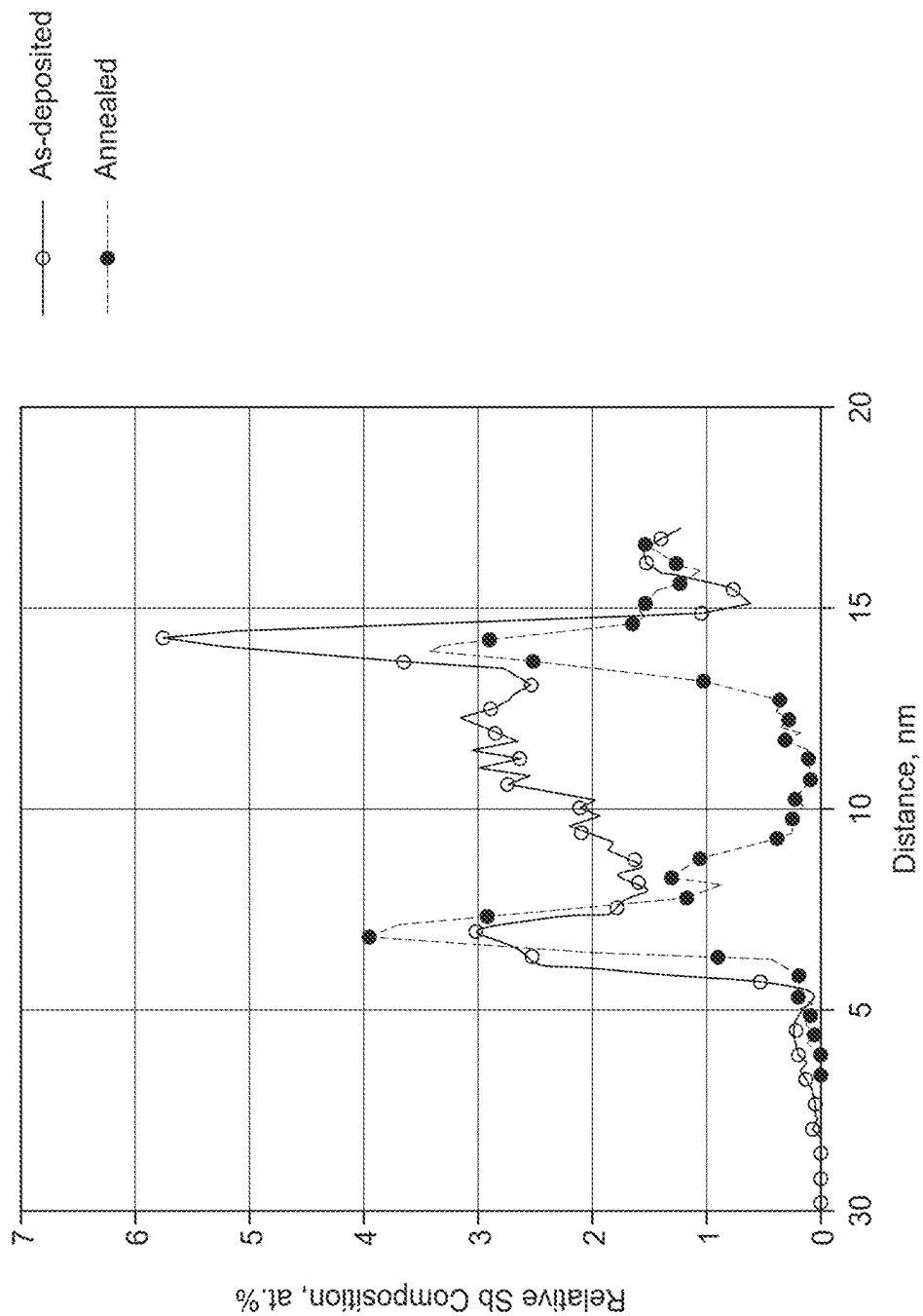


FIG. 11



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# **BUFFER LAYERS AND INTERLAYERS THAT PROMOTE BiSbx (012) ALLOY ORIENTATION FOR SOT AND MRAM DEVICES**

## **CROSS-REFERENCE TO RELATED APPLICATIONS**

This application is a continuation of co-pending U.S. patent application Ser. No. 17/401,856, filed Aug. 13, 2021, which is herein incorporated by reference.

## **BACKGROUND OF THE DISCLOSURE**

### **Field of the Disclosure**

Embodiments of the present disclosure generally relate to a buffer layer and an interlayer that inhibit antimony (Sb) migration within a bismuth antimony (BiSb) layer having a (012) orientation.

### **Description of the Related Art**

BiSb is a material that has been proposed as a spin Hall layer for spin torque oscillator (STO) and magnetoresistive random access memory (MRAM) devices. BiSb is a narrow gap topological insulator with both giant spin Hall effect and high electrical conductivity.

N. H. D. Khang, Y. Ueda, and P. N. Hai, "A conductive topological insulator with large spin Hall effect for ultralow power spin-orbit torque switching," *Nature Materials*, v. 17, 808 (2018), discovered that BiSb with a (012) crystallographic orientation has a high spin Hall angle and high conductivity in comparison to BiSb with a (001) crystallographic orientation. BiSb with a (012) crystallographic orientation was formed on a MnGa film with a (001) crystallographic orientation which was formed on a GaAs substrate with a (001) crystallographic orientation.

N. Roschewsky, E. S. Walker, P. Gowtham, S. Muschinske, F. Hellman, S. R. Bank, and S. Salahuddin, "Spin-orbit torque and Nernst effect in Bi—Sb/Co heterostructures", *Phys. Rev. B*, vol. 99, 195103 (2 May 2019), recognized that BiSb growth, crystallographic orientation, spin Hall angle, and high conductivity had poor consistency among experiments.

E. S. Walker, S. Muschinske, C. J. Brennan, S. R. Na, T. Trivedi, S. D. March, Y. Sun, T. Yang, A. Yau, D. Jung, A. F. Briggs, E. M. Krivoy, M. L. Lee, K. M. Liechti, E. T. Yu, D. Akinwande, and S. R. Bank, "Composition-dependent structural transition in epitaxial Bi1-xSbx thin films on Si(111)", *Phys. Rev. Materials* 3, 064201 (7 Jun. 2019), established growing Bi1-xSbx thin films at any thickness (including very ultra-thin films) on Si(111) substrates but only for concentrations in the 9% to 28% Sb concentration range, which happen to overlap the range needed to exhibit TI (Topological Insulator) properties. Furthermore, ultra-thin <20 Å Bi films could be grown with a strong (012) orientation, suggesting ultra-thin Bi/BiSb film laminates could be grown epitaxially with strong (012) orientation.

FIG. 11 illustrates a TEM-EELS line scan of relative Sb concentration in a 100 Å thick BiSb layer within a SOT stack without proper adjacent buffer and interlayers. FIG. 11 shows the problem of Sb migration to the interfaces from the bulk which could be improved with the use of ultra-thin Bi layers of thickness  $t$ ,  $0 < t < 10$  Å, sandwiching BiSb SOT layers that can serve as Sb composition modulations layers, to help improve the chemical uniformity and maintain (012)

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texture and structure of the BiSb layer degraded by Sb migration. However, both thin Bi and BiSb film adhesion of the BiSb layer with a (012) orientation on Si (111) was poor.

Therefore, there is a need for an improved process to form BiSb with high spin Hall angle and high conductivity and for improved devices having a BiSb layer with high spin Hall angle and high conductivity.

## **SUMMARY OF THE DISCLOSURE**

The present disclosure generally relate to spin-orbit torque (SOT) magnetic tunnel junction (MTJ) devices comprising a buffer layer, a bismuth antimony (BiSb) layer having a (012) orientation disposed on the buffer layer, and an interlayer disposed on the BiSb layer. The buffer layer and the interlayer may each independently be a single layer of material or a multilayer of material. The buffer layer and the interlayer each comprise at least one of a covalently bonded amorphous material, a tetragonal (001) material, a tetragonal (110) material, a body-centered cubic (bcc) (100) material, a face-centered cubic (fcc) (100) material, a textured bcc (100) material, a textured fcc (100) material, a textured (100) material, or an amorphous metallic material. The buffer layer and the interlayer inhibit antimony (Sb) migration within the BiSb layer and enhance uniformity of the BiSb layer while further promoting the (012) orientation of the BiSb layer.

In one embodiment, a spin-orbit torque (SOT) magnetic tunnel junction (MTJ) device comprises a substrate, a buffer layer formed over the substrate, the buffer layer comprising: an amorphous layer comprising a material in an amorphous structure, wherein the material comprises a covalently bonded carbide, a covalently bonded oxide, or a covalently bonded nitride, and a bismuth antimony (BiSb) layer formed over the buffer layer, the BiSb layer having a (012) orientation, wherein the buffer layer is configured to reduce migration of Sb in the BiSb layer.

In another embodiment, a SOT MTJ device comprises a substrate, a buffer layer formed on the substrate, the buffer layer comprising: at least one first intermediary layer, the at least one first intermediary layer comprising at least one of: a tetragonal (001) material, a tetragonal (110) material, a body-centered cubic (bcc) (100) material, a face-centered cubic (fcc) (100) material, a textured bcc (100) material, a textured fcc (100) material, a textured (100) material, or an amorphous material comprising a covalently bonded carbide, a covalently bonded oxide, or a covalently bonded nitride, and a bismuth antimony (BiSb) layer stack formed over the buffer layer comprising a BiSb layer having a (012) orientation, wherein the BiSb layer stack further comprises: a first Bi layer, wherein the BiSb layer is disposed on the first Bi layer, and a second Bi layer disposed on the BiSb layer, wherein the first and second Bi layers: each has a thickness greater than about 0 Å and less than about 10 Å, and sandwich the BiSb layer to promote a (012) BiSb texture and serve as Sb composition modulations layers configured to improve a chemical uniformity and structure of the BiSb layer degraded by Sb migration.

In yet another embodiment, a SOT MTJ device comprises a substrate and a buffer layer formed over the substrate, the buffer layer comprising: a textured layer with a (100) orientation and a first intermediary layer disposed over the textured layer, the first intermediary layer comprising at least one of a cubic crystal structure selected from the group consisting of tetragonal (001), tetragonal (110), body-centered cubic (bcc) (100), face-centered cubic (fcc) (100), textured bcc (100), and textured fcc (100). The SOT MTJ

device further comprises a bismuth antimony (BiSb) layer formed over the buffer layer, the BiSb layer having a (012) orientation, wherein the buffer layer is configured to reduce diffusion of Sb in the BiSb layer, and an interlayer disposed on the BiSb layer.

#### BRIEF DESCRIPTION OF THE DRAWINGS

So that the manner in which the above recited features of the present disclosure can be understood in detail, a more particular description of the disclosure, briefly summarized above, may be had by reference to embodiments, some of which are illustrated in the appended drawings. It is to be noted, however, that the appended drawings illustrate only typical embodiments of this disclosure and are therefore not to be considered limiting of its scope, for the disclosure may admit to other equally effective embodiments.

FIG. 1 is a schematic illustration of certain embodiments of a magnetic media drive including a MAMR write head having a SOT MTJ device.

FIG. 2 is a fragmented, cross-sectional side view of certain embodiments of a read/write head having a SOT MTJ device.

FIGS. 3A-3B illustrate spin-orbit torque (SOT) magnetic tunnel junction (MTJ) devices, according to various embodiments.

FIGS. 4A-4D illustrate exemplary multilayer structures of the buffer layer and/or the interlayer that may be utilized with the SOT MTJ devices of FIGS. 3A-3B, according to various embodiments.

FIG. 5 is a schematic cross-sectional view of a BiSb layer comprising sublayers, which may be the BiSb layer of the SOT MTJ devices of FIGS. 3A-3B, according to one embodiment.

FIGS. 6A-6E are schematic views illustrating atom lattice structures of layers within the SOT MTJ devices of FIGS. 3A-3B, according to various embodiments.

FIGS. 7-8 are graphs of a textured layer of 30 Å of RuAl in a B2 phase that is disposed on an amorphous layer of 30 Å of NiFeTa, in which a (100) fcc textured layer (MgO), a (100) bcc textured layers (Cr, Ta, W, W/Ta), and (100) B2 texture layer (NiAl) are grown on.

FIG. 9A is a schematic cross-sectional view of a SOT device for use in a MAMR write head, such as the MAMR write head of the drive of FIG. 1 or other suitable magnetic media drives.

FIGS. 9B-9C are schematic MFS views of certain embodiments of a portion of a MAMR write head with a SOT device of FIG. 9A.

FIG. 10 is a schematic cross-sectional view of a SOT MTJ used as a MRAM device.

FIG. 11 shows the problem of Sb migration to the interfaces from the bulk BiSb layer, according to one embodiment.

To facilitate understanding, identical reference numerals have been used, where possible, to designate identical elements that are common to the figures. It is contemplated that elements disclosed in one embodiment may be beneficially utilized on other embodiments without specific recitation.

#### DETAILED DESCRIPTION

In the following, reference is made to embodiments of the disclosure. However, it should be understood that the disclosure is not limited to specific described embodiments. Instead, any combination of the following features and

elements, whether related to different embodiments or not, is contemplated to implement and practice the disclosure. Furthermore, although embodiments of the disclosure may achieve advantages over other possible solutions and/or over the prior art, whether or not a particular advantage is achieved by a given embodiment is not limiting of the disclosure. Thus, the following aspects, features, embodiments and advantages are merely illustrative and are not considered elements or limitations of the appended claims except where explicitly recited in a claim(s). Likewise, reference to "the disclosure" shall not be construed as a generalization of any inventive subject matter disclosed herein and shall not be considered to be an element or limitation of the appended claims except where explicitly recited in a claim(s).

The present disclosure generally relate to spin-orbit torque (SOT) magnetic tunnel junction (MTJ) devices comprising a buffer layer, a bismuth antimony (BiSb) layer having a (012) orientation disposed on the buffer layer, and an interlayer disposed on the BiSb layer. The buffer layer and the interlayer may each independently be a single layer of material or a multilayer of material. The buffer layer and the interlayer each comprise at least one of a covalently bonded amorphous material, a tetragonal (001) material, a tetragonal (110) material, a body-centered cubic (bcc) (100) material, a face-centered cubic (fcc) (100) material, a textured bcc (100) material, a textured fcc (100) material, a textured (100) material, or an amorphous metallic material. The buffer layer and the interlayer inhibit antimony (Sb) migration within the BiSb layer and enhance uniformity of the BiSb layer while further promoting the (012) orientation of the BiSb layer.

Embodiments of the present disclosure generally relate to a buffer layer that promotes preservation of a bismuth antimony (BiSb) layer having a (012) orientation. Antimony (Sb) is highly reactive, and the buffer layer provides a low-reactive medium that reduces chemical interaction between the BiSb layer and external materials while promoting the growth of the BiSb in a (012) orientation. The configuration of the buffer layer reduces the migration of Sb in the BiSb layer.

A BiSb layer having a (012) orientation has a large spin Hall angle effect and high electrical conductivity. A BiSb layer having a (012) orientation can be used to form a spin-orbit torque (SOT) magnetic tunnel junction (MTJ) device. For example, a BiSb layer having a (012) orientation can be used as a spin Hall layer in a spin-orbit torque device in a magnetic recording head, e.g., as part of a read head, and/or a microwave assisted magnetic recording (MAMR) write head. In another example, a BiSb layer having a (012) orientation can be used as a spin Hall electrode layer in a magnetoresistive random access memory (MRAM) device. The SOT MTJ device can be in a perpendicular stack configuration or an in-plane stack configuration. The SOT MTJ device can be utilized in, for example, MAMR writing heads, in MRAM, in artificial intelligence chips, and in other applications. A BiSb layer stack **304** with a (012) orientation has a higher spin Hall angle and higher performance in a SOT MTJ device than a BiSb layer with a (001) orientation.

FIG. 1 is a schematic illustration of certain embodiments of a magnetic media drive **100** including a MAMR write head having a SOT MTJ device. Such a magnetic media drive may be a single drive or comprise multiple drives. For the sake of illustration, a single disk drive **100** is shown according to certain embodiments. As shown, at least one rotatable magnetic disk **112** is supported on a spindle **114** and rotated by a drive motor **118**. The magnetic recording on

each magnetic disk **112** is in the form of any suitable patterns of data tracks, such as annular patterns of concentric data tracks (not shown) on the magnetic disk **112**.

At least one slider **113** is positioned near the magnetic disk **112**, each slider **113** supporting one or more magnetic head assemblies **121** that include a SOT device. As the magnetic disk **112** rotates, the slider **113** moves radially in and out over the disk surface **122** so that the magnetic head assembly **121** may access different tracks of the magnetic disk **112** where desired data are written. Each slider **113** is attached to an actuator arm **119** by way of a suspension **115**. The suspension **115** provides a slight spring force which biases the slider **113** toward the disk surface **122**. Each actuator arm **119** is attached to an actuator means **127**. The actuator means **127** as shown in FIG. 2 may be a voice coil motor (VCM). The VCM includes a coil movable within a fixed magnetic field, the direction and speed of the coil movements being controlled by the motor current signals supplied by control unit **129**.

During operation of the disk drive **100**, the rotation of the magnetic disk **112** generates an air bearing between the slider **113** and the disk surface **122** which exerts an upward force or lift on the slider **113**. The air bearing thus counterbalances the slight spring force of suspension **115** and supports slider **113** off and slightly above the disk surface **122** by a small, substantially constant spacing during normal operation.

The various components of the disk drive **100** are controlled in operation by control signals generated by control unit **129**, such as access control signals and internal clock signals. Typically, the control unit **129** comprises logic control circuits, storage means and a microprocessor. The control unit **129** generates control signals to control various system operations such as drive motor control signals on line **123** and head position and seek control signals on line **128**. The control signals on line **128** provide the desired current profiles to optimally move and position slider **113** to the desired data track on disk **112**. Write and read signals are communicated to and from write and read heads on the assembly **121** by way of recording channel **125**.

The above description of a typical magnetic media drive and the accompanying illustration of FIG. 1 are for representation purposes only. It should be apparent that magnetic media drives may contain a large number of media, or disks, and actuators, and each actuator may support a number of sliders.

FIG. 2 is a fragmented, cross-sectional side view of certain embodiments of a read/write head **200** having a SOT MTJ device. The read/write head **200** faces a magnetic media **112**. The read/write head **200** may correspond to the magnetic head assembly **121** described in FIG. 1. The read/write head **200** includes a media facing surface (MFS) **212**, such as a gas bearing surface, facing the disk **112**, a MAMR write head **210**, and a magnetic read head **211**. As shown in FIG. 2, the magnetic media **112** moves past the MAMR write head **210** in the direction indicated by the arrow **232** and the read/write head **200** moves in the direction indicated by the arrow **234**.

In some embodiments, the magnetic read head **211** is a magnetoresistive (MR) read head that includes an MR sensing element **204** located between MR shields **S1** and **S2**. In other embodiments, the magnetic read head **211** is a magnetic tunnel junction (MTJ) read head that includes a MTJ sensing device **204** located between MR shields **S1** and **S2**. The magnetic fields of the adjacent magnetized regions in the magnetic disk **112** are detectable by the MR (or MTJ)

sensing element **204** as the recorded bits. The SOT MTJ device of various embodiments can be incorporated into the read head **211**.

The MAMR write head **210** includes a main pole **220**, a leading shield **206**, a trailing shield **240**, a spin orbital torque (SOT) device **250**, and a coil **218** that excites the main pole **220**. The coil **218** may have a "pancake" structure which winds around a back-contact between the main pole **220** and the trailing shield **240**, instead of a "helical" structure shown in FIG. 2. The SOT device **250** is formed in a gap **254** between the main pole **220** and the trailing shield **240**. The main pole **220** includes a trailing taper **242** and a leading taper **244**. The trailing taper **242** extends from a location recessed from the MFS **212** to the MFS **212**. The leading taper **244** extends from a location recessed from the MFS **212** to the MFS **212**. The trailing taper **242** and the leading taper **244** may have the same degree of taper, and the degree of taper is measured with respect to a longitudinal axis **260** of the main pole **220**. In some embodiments, the main pole **220** does not include the trailing taper **242** and the leading taper **244**. Instead, the main pole **220** includes a trailing side (not shown) and a leading side (not shown), and the trailing side and the leading side are substantially parallel. The main pole **220** may be a magnetic material, such as a FeCo alloy. The leading shield **206** and the trailing shield **240** may be a magnetic material, such as a NiFe alloy. In certain embodiments, the trailing shield **240** can include a trailing shield hot seed layer **241**. The trailing shield hot seed layer **241** can include a high moment sputter material, such as CoFeN or FeXN, where X includes at least one of Rh, Al, Ta, Zr, and Ti. In certain embodiments, the trailing shield **240** does not include a trailing shield hot seed layer.

FIGS. 3A-3B illustrate spin-orbit torque (SOT) magnetic tunnel junction (MTJ) devices **300**, **301**, according to various embodiments. The SOT MTJ devices **300**, **301** may each individually be used in the MAMR write head of the drive **100** of FIG. 1, the read/write head **200** of FIG. 2, or other suitable magnetic media drives.

FIG. 3A illustrates a SOT MTJ device **300**, according to one embodiment. The SOT MTJ device **300** comprises a substrate **302**, a buffer layer **310** disposed on the substrate **302**, a BiSb layer **304** or BiSb layer stack **304** comprising with a crystal orientation of (012) disposed on the buffer layer **310**, an interlayer **320** disposed on the BiSb layer **304**, a tunnel magnetoresistance (TMR)-like free layer **322** disposed on the interlayer **320**, and a MgO layer **324** disposed on the TMR-like free layer **322**. FIG. 3B illustrates a reverse SOT MTJ device **301**, according to one embodiment. The SOT MTJ device **301** comprises the substrate **302**, a MgO layer **324** disposed on the substrate **302**, a TMR-like free layer **322** disposed on the MgO layer **324**, the buffer layer **310** disposed on the TMR-like free layer **322**, the BiSb layer **304** comprising with a crystal orientation of (012) disposed on the buffer layer **310**, and the interlayer **320** disposed on the BiSb layer **304**. The SOT MTJ devices **300**, **301** comprise the same layers **302**, **304**, **310**, **320**, **322**, **324** in different arrangements.

The substrate **302** can be a silicon substrate or an alumina substrate. The silicon substrate **302** has a cubic structure of (111), (100), (100), or other crystal orientations. The alumina substrate **302** has a hexagonal structure with (001) orientations or with other crystal orientations or has an amorphous structure. The substrate **302** can be a bare substrate or can have one or more layers formed thereover, such as an oxide layer thermally grown or deposited thereover.

In one embodiment, the interlayer **320** may be the same material as the buffer layer **310**. For example, like shown in FIGS. **3A-3B**, the interlayer **320** and the buffer layer **310** may each individually comprise a single layer of a crystalline or amorphous material. In another example, the interlayer **320** and the buffer layer **310** may each individually comprise multiple layers of crystalline and/or amorphous materials. In another embodiment, the interlayer **320** and the buffer layer **310** each individually comprise one or more different materials.

The buffer layer **310** and the interlayer **320** may each individually be a multilayer structure, as discussed further below in FIGS. **4A-4D**. In one embodiment, the buffer layer **310** and/or the interlayer **320** is a covalently bonded amorphous layer. The covalently bonded amorphous material may comprise one of a covalently bonded carbide, a covalently bonded oxide, or a covalently bonded nitride. The covalently bonded amorphous material has a lattice constant of the crystal structure ( $a_{fcc}$ ) between about 3.5 Å and 3.71 Å, and the covalently bonded amorphous material has a nearest neighbor distance equal to about  $a_{fcc}$  divided by the square root of 3. In some configurations, the nearest neighbor distance is between about 2.0 Å to about 2.2 Å.

In some embodiments, the buffer layer **310** and the interlayer **320** each individually comprises one or more highly bonded materials such that the materials are less likely to interact with Sb or Bi in the BiSb layer **304** than ionic chemicals. As further discussed below in FIGS. **4A-4D**, the buffer layer **310** and the interlayer **320** may each individually comprise one or more materials selected from the group consisting of: a covalently bonded amorphous material, a tetragonal (001) material, a tetragonal (110) material, a body-centered cubic (bcc) (100) material, a face-centered cubic (fcc) (100) material, a textured bcc (100) material, a textured fcc (100) material, a textured (100) material, an amorphous metallic material, and a layered combination of one or more of any of the preceding materials.

FIGS. **4A-4D** illustrate exemplary multilayer structures of the buffer layer **310** and/or the interlayer **320** that may be utilized with the SOT MTJ devices **300**, **301** of FIGS. **3A-3B**, according to various embodiments. As shown in FIGS. **4A-4D**, the buffer layer **310** and the interlayer **320** may each individually comprise one or more amorphous or crystalline sublayers or intermediate layers **312**, **314**, **316**, **318**, and **326**.

The embodiments of FIGS. **4A-4D** can be used in combination with each other and are not an exclusive list of possible buffer layers **310** and/or interlayers **320**. Moreover, while each of FIGS. **4A-4D** describes both the buffer layer **310** and the interlayer **320**, the buffer layer **310** and the interlayer **320** may have different configurations or a different amount of sublayers or intermediate layers **312**, **314**, **316**, **318**, and **326**. Furthermore, the buffer layer **310** may be the single layer of a crystalline or amorphous material as discussed and shown above in FIGS. **3A-3B** and the interlayer **320** may be a multilayer structure as described below in FIGS. **4A-4D**, or vice versa.

In FIG. **4A**, the buffer layer **310** and/or the interlayer **320** comprises a first intermediate layer **312** and a second intermediate layer **314** disposed on the first intermediate layer **312**. In one embodiment, the first intermediate layer **312** comprises a metallic amorphous material and the second intermediate layer **314** comprises a tetragonal (001) or (110) material. In another embodiment, the first intermediate layer **312** comprises a metallic amorphous material and the second intermediate layer **314** comprises a textured (100) layer.

The tetragonal (001) or (110) material may have an a-axis in the range of about 4.49 Å to about 4.69 Å and a c-axis in the range of about 2.88 Å to about 3.15 Å. The tetragonal (001) or (110) material may have an a-axis lattice parameter in the range of about 4.20 Å to about 4.75 Å. The tetragonal (001) or (110) material may be selected from the group consisting of: SbO<sub>2</sub>, TiO<sub>2</sub>, IrO<sub>2</sub>, RuO<sub>2</sub>, CrO<sub>2</sub>, VO<sub>2</sub>, OsO<sub>2</sub>, RhO<sub>2</sub>, PdO<sub>2</sub>, WVO<sub>4</sub>, CrNbO<sub>4</sub>, SnO<sub>2</sub>, GeO<sub>2</sub>, and composites thereof with one or more elements selected from the group consisting of: W, Ta, and Nb.

The amorphous metallic material may be selected from the group consisting of: NiTa, NiFeTa, NiNb, NiW, NiFeW, NiFeHf, CoHfB, CoZrTa, CoFeB, NiFeB, CoB, FeB, and alloy combinations thereof with one or more elements selected from the group consisting of: Ni, Fe, Co, Zr, W, Ta, Hf, Ag, Pt, Pd, Si, Ge, Mn, Al, and Ti.

The textured (100) layer may be selected from the group consisting of: (1) RuAl, (2) Cr incorporated according to several options: (2a) deposited at a temperature greater than or equal to 250° C., (2b) in heated CrX alloys where X= Ru, Mo, W, or Ti < 10 at. %, or CrMo<sub>n</sub>, where n is about 20 at. % to about 50 at. %, (2c) in a stack of heated (e.g., to less than or equal to about 200° C.) Cr/CrMo<sub>n</sub> or CrMo<sub>n</sub>/Cr/CrMo<sub>n</sub>.

In FIG. **4B**, the buffer layer **310** and/or the interlayer **320** comprises a first intermediate layer **312**, a second intermediate layer **314** disposed on the first intermediate layer **312**, and a third intermediate layer **316** disposed on the second intermediate layer **314**. The first intermediate layer **312** comprises a metallic amorphous material and the second intermediate layer **314** comprises a textured (100) layer. In one embodiment, the third intermediate layer **316** comprises a textured bcc (100) layer. In another embodiment, the third intermediate layer **316** comprises an fcc (100) layer. In yet another embodiment, the third intermediate layer **316** comprises a tetragonal (001) layer.

The bcc (100) material may be selected from the group consisting of: V, Nb, Mo, W, Ta, WTi<sub>50</sub>, Al<sub>10</sub>Nb<sub>40</sub>Ti<sub>50</sub>, Cr, RuAl in a B2 phase, NiAl in a B2 phase, RhAl in a B2 phase, and alloy combinations thereof with one or more elements selected from the group consisting of: Ti, Al, Pd, Pt, Ni, Fe, and Cr.

The fcc (100) material may have a lattice parameter in the range of about 4.20 Å to about 4.70 Å. The fcc (100) material may be selected from the group consisting of: oxides, carbides, and nitrides of: (1) FeO, CoO, NiO, ZrO, MgO, TiO, ScN, TiN, NbN, ZrN, HfN, TaN, ScC, NbC, ZrC, HfC, TaC, and WC; (2) zinc blend cubic fcc (100) materials selected from the group consisting of: CoO, SiC, GaN, FeN, and ZnO; (3) composite combinations of (1) and (2) thereof with one or more elements selected from the group of W, Al, and Si; and (4) fcc metals selected from the group consisting of: MoZr<sub>10</sub>, MoNi<sub>20</sub>, NbZr<sub>20</sub>, and alloy combinations thereof with one or more elements selected from the group consisting of: W, Al, and Si. In other words, the fcc (100) is selected from the group consisting of: FeO, CoO, ZrO, MgO, TiO, ScN, TiN, NbN, ZrN, HfN, TaN, ScC, TiC, NbC, ZrC, HfC, TaC, WC, CoO, SiC, GaN, FeN, ZnO, MoZr<sub>10</sub>, MoNi<sub>20</sub>, NbZr<sub>20</sub>, and composite combinations thereof with one or more elements selected from the group of: W, Al, and Si.

In FIG. **4C**, the buffer layer **310** and/or the interlayer **320** comprises a first intermediate layer **312**, a second intermediate layer **314** disposed on the first intermediate layer **312**, a third intermediate layer **316** disposed on the second intermediate layer **314**, and a fourth intermediate layer **318** disposed on the third intermediate layer **316**. The first intermediate layer **312** comprises a metallic amorphous

material and the second intermediate layer **314** comprises a textured (100) layer. In one embodiment, the third intermediate layer **316** comprises a textured bcc (100) layer and the fourth intermediate layer **318** comprises an fcc (100) layer.

In another embodiment, the third intermediate layer **316** comprises a textured (100) bcc material and the fourth intermediate layer **318** comprises a tetragonal (110) material. In yet another embodiment, the third intermediate layer **316** comprises a textured (100) bcc material and the fourth intermediate layer **318** comprises a tetragonal (001) material.

In FIG. 4D, the buffer layer **310** and/or the interlayer **320** comprises a first intermediate layer **312** that comprises a metallic amorphous material, a second intermediate layer **314** that comprises a textured (100) material, a third intermediate layer **316** that comprises a textured (100) bcc material, a fourth intermediate layer **318** that comprises a tetragonal (001) material, and a fifth intermediate layer **326** that comprises an fcc (100) material.

In certain embodiments, the buffer layer **310** and/or the interlayer **320** are deposited by physical vapor deposition (PVD), such as sputtering, molecular beam epitaxy, ion beam deposition, other suitable PVD processes, or combinations thereof. In certain embodiments, the buffer layer **310** and/or the interlayer **320** are deposited at ambient temperatures, such as from 20° C. to about 25° C. In one aspect, forming the buffer layer **310** and/or the interlayer **320** at ambient temperatures reduces thermal migration of the intermediary layers **312**, **314**, **316**, **318**, and **326**. In another aspect, forming the buffer layer **310** at ambient temperatures minimizes altering the magnetization direction of magnetic materials formed on substrate **302** prior to forming the buffer layer **310**.

In certain embodiments, a post etch of the buffer layer **310** and/or the interlayer **320** is conducted. For example, the buffer layer **310** and/or the interlayer **320** can be post etched by an ion etch, such as directing argon ions to etch the intermediary layer **312**, **314**, **316**, **318**, and **326** on which the BiSb layer **304** is disposed. It is believed that a post etch enhances the interface between the intermediary layer **312**, **314**, **316**, **318**, and **326** and the BiSb layer **304** by cleaning the surface of the intermediary layer **312**, **314**, **316**, **318**, and **326** and/or by distorting the intermediary layer **312**, **314**, **316**, **318**, and **326** to promote (012) growth of the BiSb layer **304** thereover.

By including a material that matches the BiSb (012) textured surface of the BiSb layer **304**, such as at least one of a covalently bonded amorphous material, a tetragonal (001) material, a tetragonal (110) material, a body-centered cubic (bcc) (100) material, a face-centered cubic (fcc) (100) material (both rock salt and zinc blend), a textured bcc (100) material, a textured fcc (100) material, a textured (100) material, or an amorphous metallic material, in the buffer layer **310** and the interlayer **320** disposed in contact with the BiSb layer **304**, a (012) growth of the BiSb layer **304** is promoted and surface roughness of the BiSb layer **304** is reduced by reducing the overall grain size of the Bi and Sb atoms of the BiSb layer **304**. Improving or maintaining the BiSb (012) textured surface reduces chemical interactions with the BiSb layer **304**, which inhibits Sb migration of the BiSb layer **304**. Furthermore, including a material that matches the BiSb (012) textured surface of the BiSb layer **304** disposed in contact with the BiSb layer **304** improves epitaxy, reduces roughness, and enhances uniformity of the BiSb layer **304**.

FIG. 5 is a schematic cross-sectional view of a BiSb layer stack **304** comprising sublayers, which may be the BiSb

layer stack **304** of the SOT MTJ devices **300**, **301** of FIGS. 3A-3B, according to one embodiment. The BiSb layer stack **304** comprises Bi laminates **304a**, **304c**. A first Bi laminate **304a** is disposed on the buffer layer **310**. A BiSb layer **304b** is disposed on the first Bi laminate **304a**. The BiSb layer **304b** may comprise Sb in an atomic percentage of about 10% to about 20%. A second Bi laminate **304c** is disposed on the BiSb layer **304b**. In some embodiments, the first and second Bi laminates **304a**, **304c** each has a thickness of about 0 Å to about 10 Å.

The BiSb layer stack **304** has a (012) orientation. In some embodiments, the BiSb layer stack **304** comprises Bi1-xSbx wherein x is 0<x<1. In certain embodiments, the BiSb layer stack **304** comprises Bi1-xSbx wherein x is 0.05<x<0.22 or comprises antimony in an atomic percent content from about 7% to about 22%. The BiSb layer stack **304** has a thickness of about 20 Å to about 200 Å, such as about 50 Å to about 150 Å.

TABLE 1 shows one example of the properties of a BiSb layer stack **304** with a (012) orientation in comparison to beta-tantalum and a BiSb layer with a (001) orientation.

TABLE 1

|            | Spin Hall angle<br>$\theta_{SH}$ | conductivity<br>$\sigma$ ( $10^6 \Omega^{-1}\text{m}^{-1}$ ) | Power<br>(relative)   |
|------------|----------------------------------|--------------------------------------------------------------|-----------------------|
| Beta-Ta    | -0.15                            | 0.52                                                         | 1                     |
| BiSb (001) | 11                               | 0.25                                                         | $3.9 \times 10^{-04}$ |
| BiSb (012) | 52                               | 0.25                                                         | $1.7 \times 10^{-05}$ |

A BiSb layer stack **304** with a (012) orientation has similar electrical conductivity to the beta-tantalum (Beta-Ta) and a much larger spin Hall angle than Beta-Ta or a BiSb layer with a (001) orientation. Therefore, the relative power to produce a spin Hall effect is lower for BiSb (012) in comparison to Beta-Ta or BiSb (001).

FIGS. 6A-6E are schematic views illustrating atom lattice structures of layers within the SOT MTJ devices **300**, **301** of FIGS. 3A-3B, according to various embodiments. The various atoms in each of FIG. 6A-6E are represented by dots.

As shown in FIG. 6A, a BiSb layer **304** or a BiSb layer stack **304** with a (012) orientation has a rectangular surface with dimensions of a=about 4.54 Å and b=about 4.75 Å with about 3% mismatch in one direction. Even though the crystalline lattice structure of the BiSb layer or layer stack **304** is rectangular, it can be approximated as a square lattice were a=4.64 Å. When the buffer layer **310** and/or the interlayer **320** are made of material with a lattice structure or a nearest neighbor distance near 4.64 Å, the buffer layer **310** material and/or the interlayer **320** material can also be used to promote the growth of BiSb in a (012) orientation.

FIG. 6B illustrates a comparison of the crystalline configuration of fcc and bcc lattice structures. The filled dots represent bcc atoms and the open dots represent fcc atoms. As shown in FIG. 6B, the  $a_{fcc}$  is equal to  $a_{bcc}\sqrt{2}$ . Thus, both the crystalline configurations of fcc and bcc lattice structures can be approximated as square lattices.

FIG. 6C is an exemplary orthogonal view illustrating the tetragonal structure of RuO<sub>2</sub> in a (001) orientation, where the filled in dots represent Ru and the open dots represent O. RuO<sub>2</sub> is one example of a tetragonal crystal lattice structure in a (001) orientation that may be utilized in the buffer layer **310** and/or the interlayer **320**. However, one skilled in the art would know that other chemicals with a tetragonal (001) orientation may be utilized instead, and the tetragonal (001) orientation is not intended to be limited to RuO<sub>2</sub>. For RuO<sub>2</sub>, the rectangular prismatic dimensions are a=4.50 Å, b=4.50

Å, and  $c=3.10$  Å. Therefore,  $\text{RuO}_2$  has a 2D square lattice structure where  $a=4.50$  Å. Compared to the crystal lattice structure of BiSb, where  $a=4.64$  Å, there is a 3% difference. Therefore,  $\text{RuO}_2$  with an (001) orientation can be utilized to grow BiSb in a (012) texture.

FIG. 6D is a schematic plan view illustrating the tetragonal structure of  $\text{RuO}_2$  in a (110) orientation. FIG. 6E illustrates a comparison of the crystalline configuration of MgO in an fcc (100) orientation and  $\text{RuO}_2$  in a tetragonal (110) orientation. For  $\text{RuO}_2$  with a (110) orientation,  $\text{RuO}_2$  has a crystal lattice structure where  $a=6.36$  Å and  $b=3.10$  Å. As seen in the dashed square in FIG. 6D, there is a larger mismatch between BiSb and  $\text{RuO}_2$  in a tetragonal (110) orientation due to the Ru—Ru and Ru—O bond length differences. There is a 37% mismatch between the BiSb  $a$  value and the  $\text{RuO}_2$   $a$  value, and a 33% mismatch between the BiSb  $a$  value and the  $\text{RuO}_2$   $b$  value. However, as seen in FIGS. 6D-6E, the distances between the oxygen atoms (represented by the open dots) matches with BiSb in a (012) orientation.

FIGS. 7 and 8 are graphs 700, 800 of a textured layer of about 30 Å of RuAl in a B2 phase that is disposed on an amorphous layer of about 30 Å of NiFeTa. In FIG. 7, the graph 700 shows that this layering configuration with a close lattice match results in a uniform crystalline lattice structure. A tenuous lattice match results in some of the material forming a crystal lattice structure that is different from the bulk of the material. In FIG. 7, either MgO or Ta is disposed on the RuAl layer. RuAl in the B2 phase has an  $a_{\text{bcc}}$  of 3.000 Å, and Ta has an  $a_{\text{bcc}}$  of 3.306 Å, which is more than a 10% difference in value. Due to this poor lattice match, some of the Ta forms in a (110) orientation and the rest forms in a (200) orientation. In contrast, the  $a\sqrt{2}$  value for RuAl in the B2 phase is 4.243 Å and the  $a_{\text{fcc}}$  of MgO is 4.210 Å. Since these values are close, the majority of the MgO forms in a (100) orientation.

Other fcc materials similar to MgO have conductive properties. Therefore, the fcc materials will reduce the shunting, improve the epitaxy, and improve the growth of the BiSb. A number of the tetragonal oxides are also conductive with high resistivities, contributing the similar improved growth properties as the fcc materials.

In FIG. 8, the graph 800 shows that layering of the materials to have a closer lattice structure results in a more uniform crystalline lattice structure growth for the BiSb layer or layer stack 304. In FIG. 8, 10 Å of W is layered on the RuAl layer, and 30 Å of Ta is layered on the W layer. RuAl in the B2 phase has an  $a_{\text{bcc}}$  of 3.000 Å, W has an  $a_{\text{bcc}}$  of 3.165 Å, and Ta has an  $a_{\text{bcc}}$  of 3.306 Å. Since the difference in  $a_{\text{bcc}}$  between each layer is less than 10%, the crystalline lattice structure is more uniform than the lattice structure formed with more than a 10% difference. The Ta (100) texture is improved by using W as a strain buffer layer. Other B2 materials with lattices close to RuAl can also be grown with a (100) texture (like NiAl, for example).

FIG. 9A is a schematic cross-sectional view of a SOT device 900 for use in a MAMR write head, such as the MAMR write head of the drive 100 of FIG. 1 or other suitable magnetic media drives. The SOT device 900 comprises a BiSb layer 304 with a (012) orientation formed over a buffer layer 310 formed over a substrate 302, such as the BiSb layer 304 and the buffer layer 310 of FIGS. 3A-5. A spin torque layer (STL) 970 is formed over the BiSb layer 304. The STL 970 comprises a ferromagnetic material such as one or more layers of CoFe, CoIr, NiFe, and CoFeX alloy wherein X=B, Ta, Re, or Ir.

In certain embodiments, an electrical current shunt block layer 960 is disposed between the BiSb layer 304 and the STL 970. The electrical current shunt blocking layer 960 reduces electrical current from flowing from the BiSb layer 304 to the STL 970 but allows spin orbital coupling of the BiSb layer 304 and the STL 970. In certain embodiments, the electrical current shunt blocking layer 960 comprises a magnetic material which provides greater spin orbital coupling between the BiSb layer 304 and the STL 970 than a non-magnetic material. In certain embodiments, the electrical current shunt blocking layer 960 comprises a magnetic material of FeCo, FeCoM, FeCoMO, FeCoMMeO, FeCoM/MeO stack, FeCoMnMnMgZnFeO, FeCoM/NiMnMgZn-FeO stack, multiple layers/stacks thereof, or combinations thereof in which M is one or more of B, Si, P, Al, Hf, Zr, Nb, Ti, Ta, Mo, Mg, Y, Cu, Cr, and Ni, and Me is Si, Al, Hf, Zr, Nb, Ti, Ta, Mg, Y, or Cr. In certain embodiments, the electrical current shunt blocking layer 960 is formed to a thickness from about 10 Å to about 100 Å. In certain aspects, an electrical current shunt blocking layer 960 having a thickness of over 100 Å may reduce spin orbital coupling of the BiSb layer 304 and the STL 970. In certain aspects, an electrical current shunt blocking layer having a thickness of less than 10 Å may not sufficiently reduce electrical current from BiSb layer 304 to the STL 970.

In certain embodiments, additional layers are formed over the STL 970 such as a spacer layer 980 and a pinning layer 990. The pinning layer 990 can partially pin the STL 970. The pinning layer 990 comprises a single or multiple layers of PtMn, NiMn, IrMn, IrMnCr, CrMnPt, FeMn, other anti-ferromagnetic materials, or combinations thereof. The spacer layer 980 comprises single or multiple layers of magnesium oxide, aluminum oxide, other non-magnetic materials, or combinations thereof.

FIGS. 9B-9C are schematic MFS views of certain embodiments of a portion of a MAMR write head 210 with a SOT device 900 of FIG. 9A. The MAMR write head 210 can be the write head FIG. 2 or other suitable write heads in the drive 100 of FIG. 1 or other suitable magnetic media drives such as tape drives. The MAMR write head 210 includes a main pole 220 and a trailing shield 240 in a track direction. The SOT device 900 is disposed in a gap between the main pole and the trailing shield 240.

During operation, charge current through a BiSb layer or layer stack 304 acting as a spin Hall layer generates a spin current in the BiSb layer. The spin orbital coupling of the BiSb layer and a spin torque layer (STL) 970 causes switching or precession of magnetization of the STL 970 by the spin orbital coupling of the spin current from the BiSb layer 304. Switching or precession of the magnetization of the STL 970 can generate an assisting AC field to the write field. Energy assisted write heads based on SOT have multiple times greater power efficiency in comparison to MAMR write heads based on spin transfer torque. As shown in FIG. 9B, an easy axis of a magnetization direction of the STL 970 is perpendicular to the MFS from shape anisotropy of the STL 970, from the pinning layer 990 of FIG. 9A, and/or from hard bias elements proximate the STL 970. As shown in FIG. 9C, an easy axis of a magnetization direction of the STL 970 is parallel to the MFS from shape anisotropy of the STL 970, from the pinning layer 990 of FIG. 9A, and/or from hard bias elements proximate the STL 970.

FIG. 10 is a schematic cross-sectional view of a SOT MTJ 1001 used as a MRAM device 1000. The MRAM device 1000 comprises a reference layer (RL) 1010, a spacer layer 1020 over the RL 1010, a recording layer 1030 over the spacer layer 1020, a buffer layer 310 over an electrical

current shunt block layer **1040** over the recording layer **1030**, and a BiSb layer or layer stack **304** over the buffer layer **310**. The BiSb layer **304** and the buffer layer **310** may be the BiSb layer **304** and the buffer layer **310** of FIGS. 3A-5.

The RL **1010** comprises single or multiple layers of CoFe, other ferromagnetic materials, and combinations thereof. The spacer layer **1020** comprises single or multiple layers of magnesium oxide, aluminum oxide, other dielectric materials, or combinations thereof. The recording layer **1030** comprises single or multiple layers of CoFe, NiFe, other ferromagnetic materials, or combinations thereof.

As noted above, in certain embodiments, the electrical current shunt block layer **1040** is disposed between the buffer layer **310** and the recording layer **1030**. The electrical current shunt blocking layer **1040** reduces electrical current from flowing from the BiSb layer **304** to the recording layer **1030** but allows spin orbital coupling of the BiSb layer **304** and the recording layer **1030**. For example, writing to the MRAM device can be enabled by the spin orbital coupling of the BiSb layer and the recording layer **1030**, which enables switching of magnetization of the recording layer **1030** by the spin orbital coupling of the spin current from the BiSb layer **304**. In certain embodiments, the electrical current shunt blocking layer **1040** comprises a magnetic material which provides greater spin orbital coupling between the BiSb layer **304** and the recording layer **1030** than a non-magnetic material. In certain embodiments, the electrical current shunt blocking layer **1040** comprises a magnetic material of FeCoM, FeCoMO, FeCoMMeO, FeCoM/MeO stack, FeCoMNiMnMgZnFeO, FeCoM/NiMnMgZnFeO stack, multiple layers/stacks thereof, or combinations thereof, in which M is one or more of B, Si, P, Al, Hf, Zr, Nb, Ti, Ta, Mo, Mg, Y, Cu, Cr, and Ni, and Me is Si, Al, Hf, Zr, Nb, Ti, Ta, Mg, Y, or Cr.

The MRAM device **1000** of FIG. **10** may include other layers, such as pinning layers, pinning structures (e.g., a synthetic antiferromagnetic (SAF) pinned structure), electrodes, gates, and other structures. Other MRAM devices besides the structure of FIG. **10** can be formed utilizing a BiSb layer **304** with a (012) orientation over a buffer layer **310** to form a SOT MTJ **1001**.

By including a material that matches the BiSb (012) textured surface of the BiSb layer, such as at least one of a covalently bonded amorphous material, a tetragonal (001) material, a tetragonal (110) material, a body-centered cubic (bcc) (100) material, a face-centered cubic (fcc) (100) material, a textured bcc (100) material, a textured fcc (100) material, a textured (100) material, or an amorphous metallic material, in the buffer layer and the interlayers disposed in contact with the BiSb layer, a (012) growth of the BiSb layer is promoted and surface roughness of the BiSb layer is reduced by reducing the overall grain size of the Bi and Sb atoms of the BiSb layer. Improving or maintaining the BiSb (012) textured surface reduces chemical interactions with the BiSb layer, which inhibits Sb migration of the BiSb layer. Furthermore, including a material that matches the BiSb (012) textured surface of the BiSb layer disposed in contact with the BiSb layer improves epitaxy, reduces roughness, and enhances uniformity of the BiSb layer.

In one embodiment, a spin-orbit torque (SOT) magnetic tunnel junction (MTJ) device comprises a substrate, a buffer layer formed over the substrate, the buffer layer comprising: an amorphous layer comprising a material in an amorphous structure, wherein the material comprises a covalently bonded carbide, a covalently bonded oxide, or a covalently bonded nitride, and a bismuth antimony (BiSb) layer formed

over the buffer layer, the BiSb layer having a (012) orientation, wherein the buffer layer is configured to reduce migration of Sb in the BiSb layer.

The buffer layer further comprises one or more sublayers, each of the one or more sublayers comprising one or more materials selected from the group consisting of: a tetragonal (001) material, a tetragonal (110) material, a body-centered cubic (bcc) (100) material, a face-centered cubic (fcc) (100) material, a textured bcc (100) material, a textured fcc (100) material, a textured (100) material, an amorphous metallic material, and a layered combination of one or more of any of the preceding materials. At least one sublayer of the one or more sublayers comprises the bcc (100) material, the bcc (100) material being selected from the group consisting of: V, Nb, Mo, W, Ta,  $\text{W}_{50}\text{Ti}_{50}$ ,  $\text{Al}_{10}\text{Nb}_{40}\text{Ti}_{50}$ , Cr, CrMo, CrX where X=Ti, W, Mo, Ru, or RuAl in a B2 phase, and alloy combinations thereof with elements selected from the group consisting of: Ti, Al, Pd, Pt, Ni, Fe, and Cr. At least one sublayer of the one or more sublayers comprises the fcc (100) material, the fcc (100) material being selected from the group consisting of: FeO, CoO, ZrO, MgO, TiO, ScN, TiN, NbN, ZrN, HfN, TaN, ScC, TiC, NbC, ZrC, HfC, TaC, WC, CoO, SiC, GaN, FeN, ZnO,  $\text{MoZr}_{10}$ ,  $\text{MoNi}_{20}$ ,  $\text{NbZr}_{20}$ , and composite combinations thereof with one or more elements selected from the group of: W, Al, and Si.

At least one sublayer of the one or more sublayers comprises the tetragonal (001) or (110) material, the tetragonal (001) or (110) material being selected from the group consisting of:  $\text{SbO}_2$ ,  $\text{TiO}_2$ ,  $\text{IrO}_2$ ,  $\text{RuO}_2$ ,  $\text{CrO}_2$ ,  $\text{VO}_2$ ,  $\text{OsO}_2$ ,  $\text{RhO}_2$ ,  $\text{PdO}_2$ ,  $\text{WVO}_4$ ,  $\text{CrNbO}_4$ ,  $\text{SnO}_2$ ,  $\text{GeO}_2$ , composites thereof, and with elements selected from the group consisting of: W, Ta and Nb. The tetragonal (001) or (110) material has an a-axis in the range of about 4.49 Å to about 4.69 Å and a c-axis in the range of about 2.88 Å to about 3.15 Å. At least one sublayer of the one or more sublayers comprises the amorphous metallic material, the amorphous metallic material being selected from the group consisting of: NiTa, NiFeTa, NiNb, NiW, NiFeW, NiFeHf, CoHfB, CoZrTa, CoFeB, NiFeB, CoB, FeB, and alloy combinations thereof with elements selected from the group consisting of: Ni, Fe, Co, Zr, W, Ta, Hf, Ag, Pt, Pd, Si, Ge, Mn, Al, and Ti.

The material in the amorphous structure has a nearest neighbor XRD diffraction peak with d-spacing equal to about 2.0 Å to about 2.2 Å, corresponding to a (111) d-spacing from  $a_{\text{fcc}}$  crystal structure where  $a_{\text{fcc}}$  is between about 3.5 Å and 3.8 Å. The SOT MTJ device further comprises an interlayer disposed on the BiSb layer, the interlayer comprising one or more materials selected from the group consisting of: a tetragonal (001) material, a tetragonal (110) material, a body-centered cubic (bcc) (100) material, a face-centered cubic (fcc) (100) material, a textured bcc (100) material, a textured fcc (100) material, a textured (100) material, an amorphous material comprising covalently bonded carbide, oxide, or nitride, an amorphous metallic material, and a layered combination of one or more of any of the preceding materials.

In another embodiment, a SOT MTJ device comprises a substrate, a buffer layer formed on the substrate, the buffer layer comprising: at least one first intermediary layer, the at least one first intermediary layer comprising at least one of: a tetragonal (001) material, a tetragonal (110) material, a body-centered cubic (bcc) (100) material, a face-centered cubic (fcc) (100) material, a textured bcc (100) material, a textured fcc (100) material, a textured (100) material, or an amorphous material comprising a covalently bonded carbide, a covalently bonded oxide, or a covalently bonded nitride, and a bismuth antimony (BiSb) layer stack formed

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over the buffer layer comprising a BiSb layer having a (012) orientation, wherein the BiSb layer stack further comprises: a first Bi layer, wherein the BiSb layer is disposed on the first Bi layer, and a second Bi layer disposed on the BiSb layer, wherein the first and second Bi layers: each has a thickness greater than about 0 Å and less than about 10 Å, and sandwich the BiSb layer to promote a (012) BiSb texture and serve as Sb composition modulations layers configured to improve a chemical uniformity and structure of the BiSb layer degraded by Sb migration.

The buffer layer further comprises an amorphous layer disposed below the first intermediary layer, the amorphous layer comprising a material selected from the group consisting of: NiTa, NiFeTa, NiNb, NiW, NiFeW, NiFeHf, CoHfB, CoZrTa, CoFeB, NiFeB, CoB, FeB, and alloy combinations thereof with elements selected from the group consisting of: Ni, Fe, Co, Zr, W, Ta, Hf, Ag, Pt, Pd, Si, Ge, Mn, Al, and Ti. The buffer layer further comprises at least one second intermediary layer. The at least one second intermediary layer is a textured bcc (100) material. The at least one second intermediary layer is a textured fcc (100) material. The SOT MTJ device further comprises an interlayer disposed on the BiSb layer, the interlayer comprising a same material as the at least one first intermediary layer. The first and second Bi layers each has a width of about 0 Å to about 10 Å. The at least one first intermediary layer comprises the tetragonal (001) or (110) material, the tetragonal (001) or (110) material having an a-axis lattice parameter in the range of about 4.18 Å to about 4.75 Å. The at least one first intermediary layer comprises the fcc (100) material, the fcc (100) material having a lattice parameter in the range of about 4.18 Å to about 4.75 Å.

In yet another embodiment, a SOT MTJ device comprises a substrate and a buffer layer formed over the substrate, the buffer layer comprising: a textured layer with a (100) orientation and a first intermediary layer disposed over the textured layer, the first intermediary layer comprising at least one of a cubic crystal structure selected from the group consisting of tetragonal (001), tetragonal (110), body-centered cubic (bcc) (100), face-centered cubic (fcc) (100), textured bcc (100), and textured fcc (100). The SOT MTJ device further comprises a bismuth antimony (BiSb) layer formed over the buffer layer, the BiSb layer having a (012) orientation, wherein the buffer layer is configured to reduce diffusion of Sb in the BiSb layer, and an interlayer disposed on the BiSb layer.

The buffer layer further comprises an amorphous layer comprising a material selected from the group consisting of: NiTa, NiFeTa, NiNb, NiW, NiFeW, NiFeHf, CoHfB, CoZrTa, CoFeB, NiFeB, CoB, FeB, and alloy combinations thereof with elements selected from the group consisting of: Ni, Fe, Co, Zr, W, Ta, Hf, Ag, Pt, Pd, Si, Ge, Mn, Al, and Ti. The amorphous layer is disposed over the substrate, the textured layer is disposed on the amorphous layer, the first intermediary layer is disposed on the textured layer, and the BiSb layer is disposed on the first intermediary layer. The buffer layer further comprises one or more second intermediary layers, the one or more second intermediary layers having a cubic crystal structure different than the first intermediary layer.

The interlayer comprises one or more materials selected from the group consisting of: a tetragonal (001) material, a tetragonal (110) material, a body-centered cubic (bcc) (100) material, a face-centered cubic (fcc) (100) material, a textured bcc (100) material, a textured fcc (100) material, a textured (100) material, an amorphous material comprising a covalently bonded carbide, a covalently bonded oxide, or

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a covalently bonded nitride, an amorphous metallic material, and a layered combination of one or more of any of the preceding materials. The textured layer is selected from the group consisting of: RuAl and Cr deposited at a temperature of about 250° C. or greater, Cr in heated CrX alloys where X=Ru, Mo, W, or Ti in less than about 10 atomic percent, the CrX alloys being heated to a temperature less than or equal to about 200° C., as CrMo<sub>n</sub>, where n is about 20 atomic percent to about 50 atomic percent, or in a sandwich of heated Cr/CrMo<sub>n</sub> or CrMo<sub>n</sub>/Cr/CrMo<sub>n</sub>, where n is about 20 atomic percent to about 50 atomic percent.

While the foregoing is directed to embodiments of the present disclosure, other and further embodiments of the disclosure may be devised without departing from the basic scope thereof, and the scope thereof is determined by the claims that follow.

What is claimed is:

1. A spin-orbit torque (SOT) device, comprising:

a substrate;

a bismuth antimony (BiSb) layer disposed over the substrate, the BiSb layer having a (012) orientation; and a multilayer structure disposed in contact with the BiSb layer, the multilayer structure comprising:

an amorphous sublayer comprising a material in an amorphous structure, wherein the material comprises a carbide, a oxide, or a nitride, and wherein the material in the amorphous sublayer has a lattice constant of a crystal structure ( $a_{fcc}$ ) between about 3.5 Å and 3.71 Å.

2. The SOT device of claim 1, wherein the multilayer structure further comprises one or more sublayers in addition to the amorphous sublayer, each of the one or more sublayers comprising one or more materials selected from the group consisting of: a tetragonal (001) material, a tetragonal (110) material, a body-centered cubic (bcc) (100) material, a face-centered cubic (fcc) (100) material, a textured bcc (100) material, a textured fcc (100) material, a textured (100) material, an amorphous metallic material, and a layered combination of one or more of any of the preceding materials.

3. The SOT device of claim 1, wherein the multilayer structure is a buffer layer or an interlayer.

4. The SOT device of claim 1, wherein the material of the amorphous sublayer of the multilayer structure has a nearest neighbor distance equal to about  $a_{fcc}$  divided by the square root of 3.

5. The SOT device of claim 1, wherein a nearest neighbor distance of the material of the amorphous sublayer of the multilayer structure is about 2.0 Å to about 2.2 Å.

6. The SOT device of claim 1, wherein the amorphous sublayer comprises one or more materials selected from the group consisting of: ScC, TiC, NbC, ZrC, HfC, TaC, FeO, CoO, ZrO, MgO, TiO, TiN, NbN, ZrN, HfN, and TaN.

7. A magnetic recording head comprising the SOT device of claim 1.

8. A magnetic recording device comprising the magnetic recording head of claim 7.

9. A magneto-resistive memory comprising the SOT device of claim 1.

10. A spin-orbit torque (SOT) device, comprising:

a substrate;

a multilayer structure disposed over the substrate, the multilayer structure comprising:

at least one first intermediary layer, the at least one first intermediary layer comprising at least one of: a tetragonal (001) material, a tetragonal (110) material, a body-centered cubic (bcc) (100) material, a face-



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centered cubic (fcc) (100) material, a textured bcc (100) material, a textured fcc (100) material, a textured (100) material, and an amorphous material comprising a carbide, an oxide, or a nitride; and  
 a bismuth antimony (BiSb) layer stack disposed in contact with the multilayer structure, the bismuth antimony (BiSb) layer stack comprising:  
 a BiSb layer having a (012) orientation,  
 a first Bi layer, wherein the BiSb layer is disposed on the first Bi layer, and  
 a second Bi layer disposed on the BiSb layer,  
 wherein the first and second Bi layers sandwich the BiSb layer.

11. The SOT device of claim 10, wherein the first and second Bi layers each has a thickness greater than about 0.1 Å and less than about 10 Å.

12. The SOT device of claim 10, wherein the multilayer structure further comprises an amorphous layer disposed below the first intermediary layer, the amorphous layer comprising a material selected from the group consisting of: NiTa, NiFeTa, NiNb, NiW, NiFeW, NiFeHf, CoHfB, CoZrTa, CoFeB, NiFeB, CoB, FeB, and alloy combinations thereof with elements selected from the group consisting of: Ni, Fe, Co, Zr, W, Ta, Hf, Ag, Pt, Pd, Si, Ge, Mn, Al, and Ti.

13. The SOT device of claim 10, wherein the multilayer structure further comprises at least one second intermediary layer comprising a textured bcc (100) material or a textured fcc (100) material.

14. The SOT device of claim 10, wherein the multilayer structure is a buffer layer or an interlayer.

15. A magnetic recording head comprising the SOT device of claim 10.

16. A magnetic recording device comprising the magnetic recording head of claim 15.

17. A magneto-resistive memory comprising the SOT device of claim 10.

18. A spin-orbit torque (SOT) device, comprising:

a substrate;

a multilayer structure disposed over the substrate, the multilayer structure comprising:

a textured layer with a (100) orientation; and

a first intermediary layer disposed over the textured layer, the first intermediary layer comprising at least one of: a tetragonal (001) material, a tetragonal (110)

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material, a body-centered cubic (bcc) (100) material, a face-centered cubic (fcc) (100) material, a textured bcc (100) material, a textured fcc (100) material, a textured (100) material, and an amorphous material comprising a carbide, an oxide, or a nitride; and  
 a bismuth antimony (BiSb) layer disposed in contact with the multilayer structure, the BiSb layer having a (012) orientation.

19. The SOT device of claim 18, wherein the multilayer structure further comprises an amorphous layer comprising a material selected from the group consisting of: NiTa, NiFeTa, NiNb, NiW, NiFeW, NiFeHf, CoHfB, CoZrTa, CoFeB, NiFeB, CoB, FeB, and alloy combinations thereof with elements selected from the group consisting of: Ni, Fe, Co, Zr, W, Ta, Hf, Ag, Pt, Pd, Si, Ge, Mn, Al, and Ti, wherein the amorphous layer is disposed over the substrate, the textured layer is disposed on the amorphous layer, the first intermediary layer is disposed on the textured layer, and the BiSb layer is disposed on the first intermediary layer.

20. The SOT device of claim 18, wherein the textured layer comprises a material selected from the group consisting of:

(A) RuAl,

(B) Cr,

(C) CrX alloys where X is Ru, Mo, W, or Ti in less than about 10 atomic percent,

(D) CrMo<sub>n</sub>, where n is about 20 atomic percent to about 50 atomic percent,

(E) a layered stack of Cr and CrMo<sub>n</sub>, where n is about 20 atomic percent to about 50 atomic percent, or

(F) a layered stack of CrMo<sub>n</sub>, Cr, and CrMo<sub>n</sub>, where n is about 20 atomic percent to about 50 atomic percent.

21. The SOT device of claim 18, wherein the multilayer structure further comprises one or more second intermediary layers, the one or more second intermediary layers having a cubic crystal structure different than the first intermediary layer.

22. A magnetic recording head comprising the SOT device of claim 18.

23. A magnetic recording device comprising the magnetic recording head of claim 22.

24. A magneto-resistive memory comprising the SOT device of claim 18.

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